

Лабораторная работа №2

Простые сети в GNS3. Анализ трафика

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Цели и задачи

- Построение простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.

2.3.1. Моделирование простейшей сети на базе коммутатора в GNS3

1. Построить в GNS3 топологию сети, состоящей из коммутатора Ethernet и двух оконечных устройств (персональных компьютеров).
2. Задать оконечным устройствам IP-адреса в сети 192.168.1.0/24. Проверить связь.

2.3.2. Анализ трафика в GNS3 посредством Wireshark

1. С помощью Wireshark захватить и проанализировать ARP-сообщения.
2. С помощью Wireshark захватить и проанализировать ICMP-сообщения.

2.3.3. Моделирование простейшей сети на базе маршрутизатора FRR в GNS3

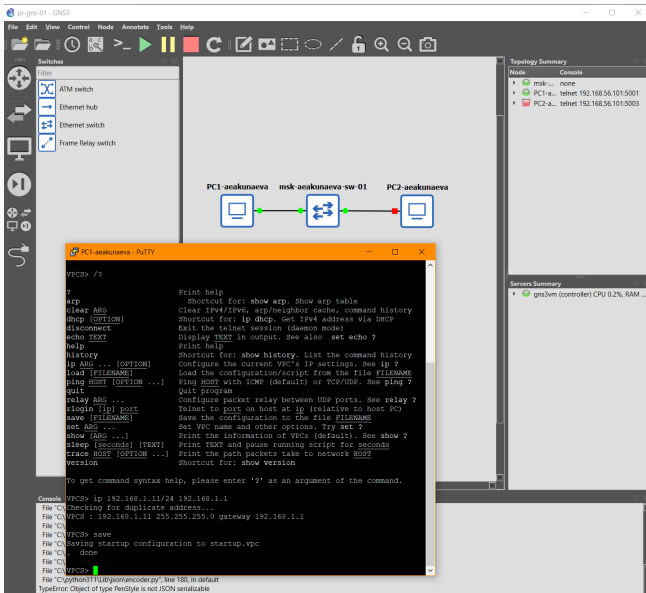
1. Построить в GNS3 топологию сети, состоящей из маршрутизатора FRR, коммутатора

Материалы и методы

- Linux Fedora Workstation (Markdown)
- Oracle VirtualBox 7.2.16
- GNS3 VM 3.0.6
- GNS3 3.0.6
- Wireshark

Выполнение лабораторной работы

Тология простейшей сети в GNS3 и задание адреса PC-1



The screenshot displays the GNS3 interface with a topology consisting of three nodes: PC1-aeakunaeva, msk-aeakunaeva-sw-01, and PC2-aeakunaeva. A terminal window for PC1-aeakunaeva is open, showing the following commands and output:

```
VPCS> /?
?                Print help
arp              Shortcut for: show arp. Show arp table
clear ARG        Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION]    Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect       Exit the telnet session (daemon mode)
echo TEXT        Display TEXT in output. See also set echo ?
help            Print help
history          Shortcut for: show history. List the command history
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?
load [FILENAME]  Load the configuration/script from the file FILENAME
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit            Quit program
relay ARG ...    Configure packet relay between UDP ports. See relay ?
rlogin [IP] PORT Telnet to port on host at ip (relative to host PC)
save [FILENAME]  Save the configuration to the file FILENAME
set ARG ...      Set VPC name and other options. Try set ?
show [ARG ...]   Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version          Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.
```

The console output shows the following commands and results:

```
Console VPCS> ip 192.168.1.11/24 192.168.1.1
File "C:\...Checking for duplicate address...
File "C:\... VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1
File "C:\...
File "C:\...VPCS> save
File "C:\...Saving startup configuration to startup.vpc
File "C:\... done
File "C:\...
File "C:\...VPCS>
```

The topology summary on the right shows the following nodes and their console output:

Node	Console
msk-...	none
PC1-...	telnet 192.168.56.101:5001
PC2-...	telnet 192.168.56.101:5003

The servers summary on the right shows the following servers:

Servers Summary
gns3vm (controller) CPU 0.2%, RAM ...

Задание адреса PC-2

PC1-aeakunaeva msk-aeakunaeva-sw-01 PC2-aeakunaeva



PC2-aeakunaeva - PuTTY

```
Dedicated to Daling.  
Build time: Sep  9 2023 11:15:00  
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)  
All rights reserved.
```

```
VPCS is free software, distributed under the terms of the "BSD" licence.  
Source code and license can be found at vpcs.sf.net.  
For more information, please visit wiki.freecode.com.cn.
```

```
Press '?' to get help.
```

```
Executing the startup file
```

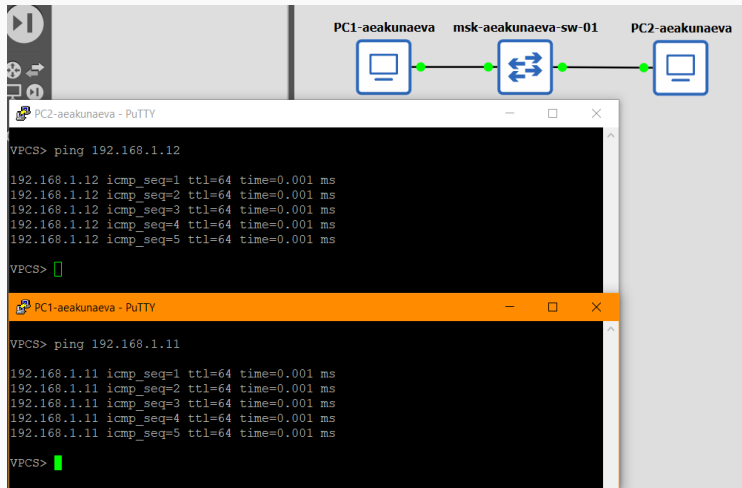
```
Hostname is too long. (Maximum 12 characters)
```

```
VPCS> ip 192.168.1.12/24 192.168.1.1  
Checking for duplicate address...  
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1
```

```
VPCS> save  
Saving startup configuration to startup.vpc  
. done
```

```
VPCS> 
```

Проверка соединений



Анализатор трафика Wireshark

The image shows the Wireshark network protocol analyzer interface. At the top, a network diagram illustrates a topology with three nodes: PC1-aeakunaeva, msk-aeakunaeva-sw-01, and PC2-aeakunaeva, connected in a linear fashion. Below the diagram, the main window displays a packet capture titled 'Захват с Standard input [msk-aeakunaeva-sw-01 Ethernet0 to PC1-aeakunaeva Ethernet0]'. The interface includes a menu bar (Файл, Правка, Вид, Запуск, Захват, Анализ, Статистика, Телефония, Беспроводная связь, Инструменты, Справка) and a toolbar. A filter bar shows 'Примените фильтр отображения ... <Ctrl-/>'. The packet list pane shows a table of captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.003475	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.050305	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
4	0.053605	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.051044	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.054102	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.051404	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.054997	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)

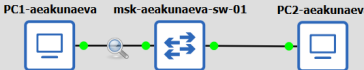
Below the packet list, the packet details pane shows the selected packet (Frame 1) with the following information:

- Frame 1: Packet, 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface 0
- Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: IPv6mcst_02 (01:00:5e:00:00:02)
- Internet Protocol Version 6, Src: ::, Dst: ff02::2
- Internet Control Message Protocol v6

The packet bytes pane shows the raw data of the selected packet in hexadecimal and ASCII format.

At the bottom of the interface, the status bar indicates 'Standard input: <live capture in progress>', 'Пакеты: 8', and 'Профиль: Default'.

Отправление эхо-запросов в разных модах



PC1-aeakunaeva msk-aeakunaeva-sw-01 PC2-aeakunaeva

PC2-aeakunaeva - PuTTY

```
1 - ICMP (default), 17 - UDP, 6 - TCP
-p port      Destination port
-s port      Source port
-T ttl       Set ttl, default 64
-t           Send packets until interrupted by Ctrl+C
-w ms        Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.
```

VPCS> ping 192.168.1.12

```
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

VPCS> ping 192.168.1.12 -l -c 1

```
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

VPCS> ping 192.168.1.11 -l -c 1

```
28 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.606 ms
28 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.520 ms
28 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.678 ms
28 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.847 ms
28 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.654 ms
```

Фильтрация пакетов по ARP-протоколу

Захват с Standard input [msk-aeakunaeva-sw-01 Ethernet0 to PC1-aeakunaeva Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
3	0.057990	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
4	1.000186	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
5	1.058972	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
6	2.001691	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
7	2.059942	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
8	461.565841	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
9	461.566185	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00

> Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0

> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Address Resolution Protocol (request/gratuitous ARP)

```
0000  ff ff ff ff ff 00 50  79 66 68 01 08 06 00 01
0010  08 00 06 04 00 01 00 50  79 66 68 01 c0 a8 01 0c
0020  ff ff ff ff ff c0 a8  01 0c 00 00 00 00 00 00
0030  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
```

Фильтрация пакетов по ICMP-протоколу

Захват с Standard input [msk-aeakunaeva-sw-01 Ethernet0 to PC1-aeakunaeva Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

icmp

No.	Time	Source	Destination	Protocol	Length	Info
10	461.567206	192.168.1.12	192.168.1.11	ICMP	42	Echo (ping) request id=0x8c0e, seq=1/256, ttl=64 (reply in 11)
11	461.567499	192.168.1.11	192.168.1.12	ICMP	42	Echo (ping) reply id=0x8c0e, seq=1/256, ttl=64 (request in 10)
12	462.570655	192.168.1.12	192.168.1.11	ICMP	42	Echo (ping) request id=0x8d0e, seq=2/512, ttl=64 (reply in 13)
13	462.570918	192.168.1.11	192.168.1.12	ICMP	42	Echo (ping) reply id=0x8d0e, seq=2/512, ttl=64 (request in 12)
14	463.572018	192.168.1.12	192.168.1.11	ICMP	42	Echo (ping) request id=0x8e0e, seq=3/768, ttl=64 (reply in 15)
15	463.572420	192.168.1.11	192.168.1.12	ICMP	42	Echo (ping) reply id=0x8e0e, seq=3/768, ttl=64 (request in 14)
16	464.573283	192.168.1.12	192.168.1.11	ICMP	42	Echo (ping) request id=0x8f0e, seq=4/1024, ttl=64 (reply in 17)
17	464.573743	192.168.1.11	192.168.1.12	ICMP	42	Echo (ping) reply id=0x8f0e, seq=4/1024, ttl=64 (request in 16)
18	465.574665	192.168.1.12	192.168.1.11	ICMP	42	Echo (ping) request id=0x900e, seq=5/1280, ttl=64 (reply in 19)
19	465.574963	192.168.1.11	192.168.1.12	ICMP	42	Echo (ping) reply id=0x900e, seq=5/1280, ttl=64 (request in 18)

> Frame 10: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0

> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:01 (00:50:79:66:68:01)

> Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11

> Internet Control Message Protocol

0000 00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00 ..Pyfh..P yfh...E-
0010 00 1c 0e 8c 00 00 40 01 e8 ed c0 a8 01 0c c0 a8@.....
0020 01 0b 08 00 6b f0 8c 0e 00 01k... ..

Фильтрация пакетов по UDP-протоколу

Захват с Standard input [msk-aekunaeva-sw-01 Ethernet0 to PC1-aekunaeva Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

udp

No.	Time	Source	Destination	Protocol	Length	Info
32	608.741705	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
33	608.741963	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
34	608.742203	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
35	608.742407	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
36	608.742704	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
37	608.742924	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
38	608.743142	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
39	608.743331	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
40	608.743517	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
41	608.743797	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
42	608.744004	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
43	608.744206	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
44	608.744404	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
45	608.744672	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0
46	608.744941	192.168.1.12	192.168.1.11	UDP	42	54105 → 7 Len=0
47	608.745159	192.168.1.11	192.168.1.12	UDP	42	7 → 54105 Len=0

< Frame 32: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:01 (00:50:79:66:68:01), Length: 42
> Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
> User Datagram Protocol, Src Port: 54105, Dst Port: 7

0000 00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00 ..Pyfh..P yfh...E
0010 00 1c 0f 1f 00 00 40 11 e8 4a c0 a8 01 0c c0 a8@..:J.....
0020 01 0b d3 59 00 07 00 08 a9 15 ...Y.....

PC1-aekunaeva - PuTTY

```
VPCS> ping 192.168.1.11 -l -c -2
192.168.1.11 udp_seq=1 timeout
192.168.1.11 udp_seq=2 timeout
192.168.1.11 udp_seq=3 timeout
192.168.1.11 udp_seq=4 timeout
192.168.1.11 udp_seq=5 timeout

VPCS> ping 192.168.1.11 -l -c -3
Connect 70192.168.1.11 seq=1 ttl=64 time=1.085 ms
SendData 70192.168.1.11 seq=1 ttl=64 time=1.044 ms
Close 70192.168.1.11 seq=1 ttl=64 time=2.087 ms
Connect 70192.168.1.11 seq=2 ttl=64 time=1.098 ms
SendData 70192.168.1.11 seq=2 ttl=64 time=1.083 ms
Close 70192.168.1.11 seq=2 ttl=64 time=2.137 ms
Connect 70192.168.1.11 seq=3 ttl=64 time=1.103 ms
SendData 70192.168.1.11 seq=3 ttl=64 time=1.032 ms
Close 70192.168.1.11 seq=3 ttl=64 time=2.282 ms
Connect 70192.168.1.11 seq=4 ttl=64 time=1.194 ms
SendData 70192.168.1.11 seq=4 ttl=64 time=1.081 ms
Close 70192.168.1.11 seq=4 ttl=64 time=2.131 ms
Connect 70192.168.1.11 seq=5 ttl=64 time=1.132 ms
SendData 70192.168.1.11 seq=5 ttl=64 time=1.065 ms
Close 70192.168.1.11 seq=5 ttl=64 time=2.179 ms
```

User Datagram Protocol: Protocol

File "C:\Users\circle\project\Windows\gn3-gui\gn3\http_client.py", line 378
File "C:\python311\Lib\json_init_.py", line 231, in dumps
File "C:\python311\Lib\json\encoder.py", line 200, in encode
File "C:\python311\Lib\json\encoder.py", line 258, in iterencode
File "C:\python311\Lib\json\encoder.py", line 180, in default
TypeError: Object of type PenStyle is not JSON serializable

```
ping
*** Unknown syntax: ping
-> ping
*** Unknown syntax: ping
```

Фильтрация пакетов по TCP-протоколу

Захват с Standard input [msk-asekunaeva-sw-01 Ethernet0 to PC1-asekunaeva Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

tcp

No.	Time	Source	Destination	Protocol	Length	Info
237301	-53.607181	192.168.1.12	192.168.1.11	TCP	74	62972 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1790447407 TSecr=0
237307	-53.606978	192.168.1.11	192.168.1.12	TCP	54	7 → 62972 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
237332	-53.605978	192.168.1.12	192.168.1.11	TCP	66	62972 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1790447407 TSecr=0
237333	-53.605903	192.168.1.12	192.168.1.11	TCP	66	[TCP Dup ACK 237332#1] 62972 → 7 [PSH, ACK] Seq=1 Ack=1 Win=2920 Len=0
237339	-53.605729	192.168.1.11	192.168.1.12	TCP	54	[TCP Dup ACK 237307#1] 7 → 62972 [ACK] Seq=1 Ack=1 Win=2920 Len=0
237360	-53.604870	192.168.1.12	192.168.1.11	TCP	66	62972 → 7 [FIN, PSH, ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1790447407 TSecr=0
237367	-53.604629	192.168.1.11	192.168.1.12	TCP	54	7 → 62972 [ACK] Seq=1 Ack=2 Win=2920 Len=0
237368	-53.604608	192.168.1.11	192.168.1.12	TCP	54	7 → 62972 [FIN, ACK] Seq=1 Ack=2 Win=2920 Len=0
237409	-53.602735	192.168.1.12	192.168.1.11	TCP	66	62972 → 7 [ACK] Seq=2 Ack=2 Win=2920 Len=0 TSval=1790447407 TSecr=0
256997	-52.602577	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 62972 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460
257001	-52.602372	192.168.1.11	192.168.1.12	TCP	54	7 → 62972 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
257019	-52.601476	192.168.1.12	192.168.1.11	TCP	66	62972 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1790447408 TSecr=0
257025	-52.601166	192.168.1.12	192.168.1.11	TCP	66	[TCP Dup ACK 257019#1] 62972 → 7 [PSH, ACK] Seq=1 Ack=1 Win=2920 Len=0
257031	-52.600985	192.168.1.11	192.168.1.12	TCP	54	[TCP Dup ACK 257001#1] 7 → 62972 [ACK] Seq=1 Ack=1 Win=2920 Len=0
257052	-52.600117	192.168.1.12	192.168.1.11	TCP	66	62972 → 7 [FIN, PSH, ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1790447408 TSecr=0
257058	-52.599926	192.168.1.11	192.168.1.12	TCP	54	7 → 62972 [ACK] Seq=1 Ack=2 Win=2920 Len=0

< >

> Frame 237301: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592) on interface 0

> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:01 (00:50:79:66:68:01)

> Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11

> Transmission Control Protocol, Src Port: 62972, Dst Port: 7, Seq: 0, Len: 0

```
0000  00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00  Pyfh...P yfh...E-
0010  00 3c 0f 2f 00 00 40 06 e8 25 c0 a8 01 0c c0 a8  <./-@.X.....
0020  01 0b f5 fc 00 07 6f a1 52 2f 00 00 00 a0 02  .....o R/.....
0030  0b 68 8a 7b 00 00 02 04 05 b4 01 01 08 0a 6a b8  .h{.....j.....
0040  0f 2f 00 00 00 00 01 03 03 01  ..../.....
```

Проект 2 в GNS3

The screenshot displays the GNS3 interface with a network topology and two terminal windows.

Topology: A diagram showing three nodes connected in a line: `PCI-acakunaeva` (PC icon) connected to `msk-acakunaeva-sw-01` (switch icon), which is connected to `msk-acakunaeva-gw-01` (router icon).

Terminal 1 (msk-acakunaeva-gw-01 - PuTTY): Shows the output of the startup file execution. It includes information about the Virtual PC Simulator version (0.9.3), build time (Sep 9 2023 11:15:00), and copyright (c) 2007-2015, Paul Meng (mirmahi@gmail.com). It also shows the execution of the startup file, which sets the hostname to `msk-acakunaeva-gw-01` and configures the network.

Terminal 2 (msk-acakunaeva-sw-01 - PuTTY): Shows the output of the startup file execution. It includes information about the Virtual PC Simulator version (0.9.3), build time (Sep 9 2023 11:15:00), and copyright (c) 2007-2015, Paul Meng (mirmahi@gmail.com). It also shows the execution of the startup file, which sets the hostname to `msk-acakunaeva-sw-01` and configures the network.

Packet Capture: A packet capture window is open, showing a packet capture on the interface `msk-acakunaeva-sw-01 Ethernet1 to msk-acakunaeva-gw-01 eth0`. The packet list shows a single packet (No. 1) at time 0.000000, source `ff02::2`, destination `ff02::2`, protocol `ICMPv6`, length 62, and info `Router Solicitation`. The packet details show the Ethernet II frame, Internet Protocol Version 6, and Internet Control Message Protocol (ICMPv6) Router Solicitation.

Настройка IP-адресации устройств

Left Terminal (VPCS):

```
Hostname is too long. (Maximum 12 characters)
VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IF/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10004
RHOST:PORT : 127.0.0.1:10005
MTU        : 1500

VPCS>
```

Right Terminal (frr):

```
frr# configure terminal
frr(config)# hostname msk-aekunaeva-gw-01
msk-aekunaeva-gw-01(config)# exit
msk-aekunaeva-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]

msk-aekunaeva-gw-01# configure terminal
msk-aekunaeva-gw-01(config)# interface eth0
msk-aekunaeva-gw-01(config-if)# ip address 192.168.1.1/24
msk-aekunaeva-gw-01(config-if)# no shutdown
msk-aekunaeva-gw-01(config-if)# exit
msk-aekunaeva-gw-01(config)# exit
msk-aekunaeva-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]

msk-aekunaeva-gw-01# show running-config
Building configuration...

Current configuration:

frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-aekunaeva-gw-01
service integrated-vtysh-config

interface eth0
vrf up address 192.168.1.1/24
exit
end
msk-aekunaeva-gw-01# show interface brief
Interface      Status    VRF        Addresses
-----
eth0           up        default    192.168.1.1/24
eth1           down      default
eth2           down      default
eth3           down      default
eth4           down      default
eth5           down      default
eth6           down      default
eth7           down      default
lo             up        default
pimreg        up        default

msk-aekunaeva-gw-01#
```

Bottom Window (Packet Capture):

Frame	Time	Source	Destination	Protocol
1	0.000000	192.168.1.10	192.168.1.1	ICMP
2	113.777341	192.168.1.1	192.168.1.10	ICMP
3	114.777361	192.168.1.1	192.168.1.10	ICMP
4	115.777751	192.168.1.1	192.168.1.10	ICMP
5	252.880188	192.168.1.10	192.168.1.1	ICMP
6	252.920123	192.168.1.10	192.168.1.1	ICMP
7	253.101562	192.168.1.10	192.168.1.1	ICMP
8	254.133070	192.168.1.10	192.168.1.1	ICMP
9	254.161283	192.168.1.10	192.168.1.1	ICMP
10	254.660049	192.168.1.1	192.168.1.10	ICMP
11	255.170222	192.168.1.1	192.168.1.10	ICMP

Проверка подключения соединения

```
PORT : 10004
RHOST:PORT : 127.0.0.1:10005
MTU : 1500

VPCS> ping 192.168.1.1

64 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.163 ms
64 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.219 ms
64 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.432 ms
64 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.327 ms

VPCS: [ ]

msk-aakunaeva-gw-01# ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1): 56 data bytes
64 bytes from 192.168.1.1: seq=0 ttl=64 time=0.432 ms
64 bytes from 192.168.1.1: seq=1 ttl=64 time=0.047 ms
64 bytes from 192.168.1.1: seq=2 ttl=64 time=0.048 ms
64 bytes from 192.168.1.1: seq=3 ttl=64 time=0.138 ms
64 bytes from 192.168.1.1: seq=4 ttl=64 time=0.318 ms
64 bytes from 192.168.1.1: seq=5 ttl=64 time=0.080 ms
64 bytes from 192.168.1.1: seq=6 ttl=64 time=0.056 ms
64 bytes from 192.168.1.1: seq=7 ttl=64 time=0.051 ms
64 bytes from 192.168.1.1: seq=8 ttl=64 time=0.045 ms
64 bytes from 192.168.1.1: seq=9 ttl=64 time=0.055 ms
64 bytes from 192.168.1.1: seq=10 ttl=64 time=0.058 ms
64 bytes from 192.168.1.1: seq=11 ttl=64 time=0.060 ms
64 bytes from 192.168.1.1: seq=12 ttl=64 time=0.051 ms
64 bytes from 192.168.1.1: seq=13 ttl=64 time=0.054 ms
64 bytes from 192.168.1.1: seq=14 ttl=64 time=0.047 ms
64 bytes from 192.168.1.1: seq=15 ttl=64 time=0.078 ms
64 bytes from 192.168.1.1: seq=16 ttl=64 time=0.051 ms
64 bytes from 192.168.1.1: seq=17 ttl=64 time=0.071 ms
64 bytes from 192.168.1.1: seq=18 ttl=64 time=0.050 ms
64 bytes from 192.168.1.1: seq=19 ttl=64 time=0.047 ms
64 bytes from 192.168.1.1: seq=20 ttl=64 time=0.049 ms
64 bytes from 192.168.1.1: seq=21 ttl=64 time=0.112 ms
64 bytes from 192.168.1.1: seq=22 ttl=64 time=0.305 ms
64 bytes from 192.168.1.1: seq=23 ttl=64 time=0.048 ms
64 bytes from 192.168.1.1: seq=24 ttl=64 time=0.054 ms
64 bytes from 192.168.1.1: seq=25 ttl=64 time=0.065 ms
64 bytes from 192.168.1.1: seq=26 ttl=64 time=0.049 ms
64 bytes from 192.168.1.1: seq=27 ttl=64 time=0.048 ms
64 bytes from 192.168.1.1: seq=28 ttl=64 time=0.052 ms
```

Захват с Standard input [msk-aakunaeva-sw-01 Ethernet1 to msk-aakunaeva-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
13	449.786136	0c:67:7a:99:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at 0c:67:7a
14	449.787288	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) request id=0x
15	449.787907	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x
16	450.788653	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) request id=0x
17	450.789356	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x
18	451.791051	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) request id=0x
19	451.791894	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x
20	452.795233	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) request id=0x
21	452.796338	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x
22	453.798040	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) request id=0x
23	453.799002	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x
24	454.837605	0c:67:7a:99:00:00	Private_66:68:00	ARP	60	Who has 192.168.1.10? Tell
25	454.838105	Private_66:68:00	0c:67:7a:99:00:00	ARP	60	192.168.1.10 is at 00:50:7

<

Frame 1: Packet, 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: IPv6mcast_02

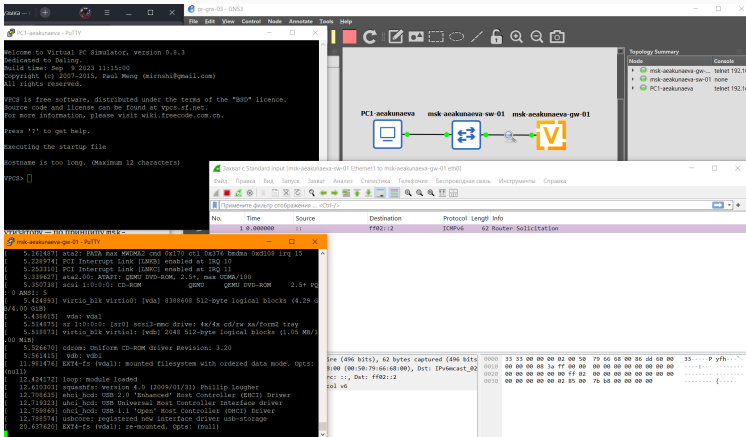
Internet Protocol Version 6, Src: ::, Dst: ff02::2

Internet Control Message Protocol v6

Standard input: <live capture in progress>

Пагинация: 25

Проект 3 в GNS3



Настройка устройств проекта 3

```
PC1-aeakunaeva - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10004
RHOST:PORT : 127.0.0.1:10005
MTU        : 1500

VPCS> █
```

```
msk-aeakunaeva-gw-01 - PuTTY
vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-aeakunaeva-gw-01
[edit]
vyos@vyos# save
Warning: you have uncommitted changes that will not be saved.

Saving configuration to '/config/config.boot'...
Done
[edit]
```

Проверка и сохранение изменений конфигурации маршрутизатора VyOS

```
msk-aeakunaeva-gw-01 - PuTTY
[edit]
vyos@vyos# del interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24

Configuration path: [interfaces ethernet eth0 address 192.168.1.1/24] already
exists

[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
- address dhcp
+ address 192.168.1.1/24
[edit system]
> host-name msk-aeakunaeva-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address 192.168.1.1/24
     hw-id 0c:f8:b2:ce:00:00
 }
 ethernet eth1 {
     hw-id 0c:f8:b2:ce:00:01
 }
 ethernet eth2 {
     hw-id 0c:f8:b2:ce:00:02
 }
 ethernet eth3 {
     hw-id 0c:f8:b2:ce:00:03
 }
 ethernet eth4 {
     hw-id 0c:f8:b2:ce:00:04
 }
 ethernet eth5 {
     hw-id 0c:f8:b2:ce:00:05
 }
 ethernet eth6 {
     hw-id 0c:f8:b2:ce:00:06
 }
 ethernet eth7 {
[edit]
vyos@vyos# exit
exit
```

Проверка соединения в Wireshark

VPCS Terminal Output:

```
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10004
RHOST:PORT : 127.0.0.1:10005
MTU        : 1500

VPCS> ping 192.168.1.1

64 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=0.697 ms
64 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.072 ms
64 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.893 ms
64 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.165 ms
64 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=0.768 ms

VPCS>
```

Network Topology Summary:

Node	Console
msk-aeskunaeva-gw-01	telnet 192
msk-aeskunaeva-sw-01	none
PC1-aeskunaeva	telnet 192

Wireshark Packet Capture (Frame 22):

No.	Time	Source	Destination	Protocol	Length	Info
28	380.095104	0c:f8:b2:ce:00:00	Private_66:68:00	ARP	42	192.168.1.1 is at 0c:f8:b2:ce:00:00
29	380.096225	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5014, seq=1/256, ttl=64 (reply in 30)
30	380.096670	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5014, seq=1/256, ttl=64 (request in 29)
31	381.097428	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5114, seq=2/512, ttl=64 (reply in 32)
32	381.098039	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5114, seq=2/512, ttl=64 (request in 31)
33	382.099374	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5214, seq=3/768, ttl=64 (reply in 34)
34	382.099992	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5214, seq=3/768, ttl=64 (request in 33)
35	383.101773	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5314, seq=4/1024, ttl=64 (reply in 36)
36	383.102282	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5314, seq=4/1024, ttl=64 (request in 35)
37	384.103979	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5414, seq=5/1280, ttl=64 (reply in 38)
38	384.104502	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5414, seq=5/1280, ttl=64 (request in 37)
39	385.335321	0c:f8:b2:ce:00:00	Private_66:68:00	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
40	385.335636	Private_66:68:00	0c:f8:b2:ce:00:00	ARP	42	192.168.1.10 is at 00:50:79:66:68:00

Frame 22 Details:

- Frame 22: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736) on interface 0
- Ethernet II, Src: 0c:f8:b2:ce:00:00 (0c:f8:b2:ce:00:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
- User Datagram Protocol, Src Port: 68, Dst Port: 67
- Dynamic Host Configuration Protocol (Discover)

Packet Bytes:

```
0000  ff ff ff ff ff ff 0c f8  b2 ce 00 00 00 45 10  .....E
0010  01 48 00 00 00 00 00 11  39 96 00 00 00 ff ff  -H.....9
0020  ff ff 00 44 00 43 81 34  87 85 01 01 06 00 68 ed  -D.C.4.....h
0030  30 14 00 09 00 00 00 00  00 00 00 00 00 00 00 00  0.....0
0040  00 00 00 00 00 00 0c f8  b2 ce 00 00 00 00 00  0.....
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  0.....
0060  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  0.....
0070  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  0.....
0080  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  0.....
0090  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  0.....
00a0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  0.....
```

Выводы

Я получила навыки построения простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.